16 y is directly proportional to $\sqrt[3]{x}$

$y = 1\frac{1}{6}$ when $x = 8$

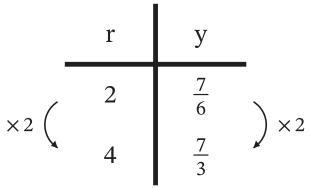
Find the value of y when $x = 64$

Let r be the cube root of x. Then y = k r.

When x = 8, r = 2: $\frac{7}{6} = 2k$ $k = \frac{7}{12}$

1 mark
Use direct proportion to the cube root

1 mark
Find the constant k



When x = 64, r = 4.

$y = \frac{7}{12} \times 4 = \frac{7}{3}$

1 mark
Correct exact value

$\frac{7}{3}$

(Total for Question 16 is 3 marks)